

Interband Absorption in Gold from Fermi's Golden Rule:

a first-principles rederivation of Rosei's ε_2 at the X and L critical points, in Rosei's own notation

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July 2026

Abstract

This document rederives, from first principles and with every prefactor tracked, the Rosei-model interband $\varepsilon_2(\hbar\omega, T)$ of gold at the X and L critical points [Guerrisi, Rosei & Winsemius (GRW), Phys. Rev. B **12**, 557 (1975)], written throughout in GRW's own notation, so that their ten numbered equations emerge explicitly from one derivation. The route is the one developed in my notes: Fermi's Golden Rule (FGR) for Bloch states in $\mathbf{k}$ -space, reduction of the transition sum to a one-dimensional energy integral through the linearization $(u, v) = (k_\perp^2, k_\parallel^2)$, and resolution of the double δ -constraint by a single 2×2 Jacobian. Seven of the ten GRW equations are recreated verbatim; the printed Eqs. (4), (6) and (8) do not come out as written, and for each the failure is stated in one line and the corrected form derived (App. A). The origin of the $1/\omega^2$ prefactor of the golden-rule ε_2 is pinned down explicitly. The photonic density of states is derived by field quantization, the spontaneous-emission counterpart is assembled, and the pair closes exactly into the van Roosbroeck–Shockley/Kirchhoff relation — the global consistency check of all prefactors. A final section gives the $k_\parallel$ quadrature scheme that evaluates the model without touching its edge singularities: the groundwork for the numerical implementation.

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1 Scope, sources and reading map

Three families of documents about interband absorption coexist in this project:

1. **My derivation chain** (appendices A1, A8, A8.L and the notes *Absorption Integral*, *Interband Absorption at X and L* , *Equivalence of My Absorption Integral and Rosei's*, and *Rebuilding Rosei's Interband Absorption in His Own Notation*): FGR in $\mathbf{k}$ -space, the (u, v) linearization, and the verified Jacobian bridge to Rosei's formulas.
2. **Attempts at recreating Rosei's own derivation** (`interband.derivation.tex`, v2, `rosei.equivalence_derivation.tex`); AI-drafted, mutually inconsistent, superseded by the present document.
3. **The primary source**, GRW 1975, whose printed Eqs. (4), (6) and (8) contain defects (App. A).

This document keeps only what survives verification, and it speaks GRW's language: bands are u (upper) and l (lower), all levels and masses are positive numbers with every sign written explicitly, and equations tagged (n) are GRW's Eq. (n) recreated — primes, as in $(6')$, mark corrected forms. Note that in the 1975 paper GRW only *derive* the X point; the L machinery is borrowed from Rosei 1974 (“will not be repeated here”). Here both points come out of the same reduction, side by side.

Section 3 builds the golden-rule machinery with all constants, including the origin of the $1/\omega^2$ prefactor; Sec. 4 performs the $\mathbf{k}$ -space reduction and lands on GRW's Eqs. $(4')$, (5) , (7) ; Sec. 5 splits into the X and L columns (Eqs. $(6')$, $(8')$); Sec. 6 assembles ε_2 (Eqs. (9) , (10)) and scores all ten equations; Sec. 7 derives the photonic density of states and the emission spectrum; Sec. 8 gives the $k_{\parallel}$ quadrature. Appendices: the GRW corrections, a dictionary of repo conventions, and a dimensional audit.

2 Conventions

Units	Gaussian throughout (GRW's system); SI conversions noted where useful.
Energy zero	Fermi level, $E_F = 0$; all electron energies E from E_F .
Bands	u = upper (conduction), l = lower (d); GRW's subscripts.
Levels & masses	$\hbar\omega_{X_6^-}, \hbar\omega_{X_7^+}, \hbar\omega_{L_6^-}, \hbar\omega_{L^+}$ and $m_{u\perp}, m_{u\parallel}, m_{l\perp}, m_{l\parallel}$ are all <i>positive numbers</i> ; curvature directions and the side of E_F are carried only by explicit $\pm$ signs, never hidden in a symbol.
Band coefficients	$A_i = \hbar^2/2m_{i\perp}$, $B_i = \hbar^2/2m_{i\parallel}$, $i = u, l$; all $A_i, B_i > 0$.
Local axes	$k_{\parallel}$ along ΓX (resp. ΓL), $k_{\perp}$ radial in the perpendicular plane; $\mathbf{k}$ measured from the critical point.
Spin	carried explicitly as a factor 2, never absorbed silently.
Counting	N counts equivalent <i>half</i> -neighborhoods ($N_X = 6$ faces, $N_L = 8$), paired with the half-space $D_{l \rightarrow u}$ below — or a factor 2 goes missing.
Occupation	Rosei's $[1 - f(E, T)]$, $f(E) = [e^{E/k_B T} + 1]^{-1}$; see Sec. 3.5.
Matrix element	$ P ^2 \equiv \langle u \nabla l \rangle ^2$ (dimension length^{-2}), GRW's convention; momentum form $ \mathbf{p}_{ul} ^2 = \hbar^2 P ^2$.
Photon	energy $\hbar\omega$, wavevector $\mathbf{q}$, polarization $\hat{\epsilon}_{\lambda}$.

3 Fermi's Golden Rule in k-space

3.1 Minimal coupling

With the radiation field described by a vector potential $\mathbf{A}$ in the Coulomb gauge ($\nabla \cdot \mathbf{A} = 0$), the one-electron Hamiltonian is

$$\hat{H} = \frac{[\hat{\mathbf{p}} + \frac{e}{c}\mathbf{A}(\hat{\mathbf{r}}, t)]^2}{2m} + V(\hat{\mathbf{r}}) = \underbrace{\frac{\hat{p}^2}{2m}}_{\hat{H}_0} + V + \underbrace{\frac{e}{2mc}(\mathbf{A} \cdot \hat{\mathbf{p}} + \hat{\mathbf{p}} \cdot \mathbf{A})}_{\hat{H}_{\text{int}}} + \underbrace{\frac{e^2 A^2}{2mc^2}}_{\text{dropped}}. \quad (3.1)$$

In the Coulomb gauge $\hat{\mathbf{p}} \cdot \mathbf{A} = \mathbf{A} \cdot \hat{\mathbf{p}}$, so

$$\boxed{\hat{H}_{\text{int}} = \frac{e}{mc} \mathbf{A} \cdot \hat{\mathbf{p}}} \quad (3.2)$$

(the $\frac{1}{2}$ of the symmetrized form is consumed by the identity of the two terms). The A^2 term does not connect different bands to first order and is dropped.

3.2 Quantized field: absorption, stimulated and spontaneous emission

Expanding the field in a quantization volume V ,

$$\mathbf{A}(\mathbf{r}) = \sum_{\mathbf{q}, \lambda} \sqrt{\frac{2\pi\hbar c^2}{\omega_q V}} \hat{\epsilon}_{\lambda} \left(\hat{a}_{\mathbf{q}\lambda} e^{i\mathbf{q} \cdot \mathbf{r}} + \hat{a}_{\mathbf{q}\lambda}^{\dagger} e^{-i\mathbf{q} \cdot \mathbf{r}} \right), \quad (3.3)$$

the matrix elements of $\hat{H}_{\text{int}}$ between joint electron-photon states carry the boson factors

$$|\langle n_q - 1 | \hat{a} | n_q \rangle|^2 = n_q \quad (\text{absorption}), \quad |\langle n_q + 1 | \hat{a}^{\dagger} | n_q \rangle|^2 = n_q + 1 \quad (\text{emission}). \quad (3.4)$$

The “+1” is the spontaneous channel: it survives at $n_q = 0$ and is invisible to a purely classical $\mathbf{A}$. The semiclassical golden rule gives absorption and *stimulated* emission only; spontaneous emission requires (3.4), and its mode sum produces the photonic density of states in Sec. 7.

3.3 Bloch states and vertical transitions

For Bloch states $\psi_{n\mathbf{k}} = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r})/\sqrt{V}$, the matrix element of $e^{\pm i\mathbf{q}\cdot\mathbf{r}} \hat{\boldsymbol{\varepsilon}} \cdot \hat{\mathbf{p}}$ enforces $\mathbf{k}' = \mathbf{k} \pm \mathbf{q}$. At optical frequencies $q \sim \omega/c \sim 10^{-3} \text{ \AA}^{-1}$ is negligible on the zone scale, so transitions are vertical ($\mathbf{k}' = \mathbf{k}$) and

$$M_{ul}(\mathbf{k}) \equiv \langle u\mathbf{k} | \hat{\boldsymbol{\varepsilon}} \cdot \hat{\mathbf{p}} | l\mathbf{k} \rangle, \quad |\langle f | \hat{H}_{\text{int}} | i \rangle|^2 = \left(\frac{e}{mc} \right)^2 \frac{2\pi\hbar c^2}{\omega V} |M_{ul}(\mathbf{k})|^2 \times \{n_q \text{ or } n_q + 1\}. \quad (3.5)$$

3.4 Golden-rule rate, ε_2 , and the origin of the $1/\omega^2$

FGR for one photon mode, then summing over $\mathbf{k}$ (factor 2 for spin, $\sum_{\mathbf{k}} \rightarrow V \int d^3k / (2\pi)^3$) with Pauli blocking, gives the photon absorption rate. Equating the absorbed power to the electromagnetic dissipation rate $P_{\text{abs}} = \frac{1}{8\pi} \omega \varepsilon_2(\omega) |\mathbf{E}_0|^2 V$ of a classical field $\mathbf{E} = \mathbf{E}_0 \cos \omega t$ yields

$$\boxed{\varepsilon_2(\omega) = \frac{4\pi^2 e^2}{m^2 \omega^2} \frac{2}{(2\pi)^3} \int_{\text{BZ}} d^3k |M_{ul}(\mathbf{k})|^2 \delta(\hbar\omega_u(\mathbf{k}) - \hbar\omega_l(\mathbf{k}) - \hbar\omega) [1 - f(\hbar\omega_u(\mathbf{k}))]} \quad (3.6)$$

per pair of bands (SI: replace $4\pi e^2 \rightarrow e^2/\varepsilon_0$). The occupation factor is Rosei's, discussed in Sec. 3.5.

Remark 3.1 (Why exactly $1/\omega^2$) *The ω -dependence of the prefactor is fixed before any band structure enters, by three elementary counts. (i) The vertex: the field couples through $\mathbf{A}$, not $\mathbf{E}$, and for a classical wave $\mathbf{E} = -\frac{1}{c}\partial_t \mathbf{A}$ means $A_0 = cE_0/\omega$. The golden-rule rate therefore carries $|eA_0/mc|^2 = e^2 E_0^2 / m^2 \omega^2$ — two inverse powers of ω from expressing the $\mathbf{A} \cdot \mathbf{p}$ coupling in terms of the measured field. (ii) Energy per event: each absorption removes one quantum, $P_{\text{abs}} = \hbar\omega \times \text{rate}$: one positive power. (iii) The definition of ε_2 : dissipation per unit field is $P/V = \omega \varepsilon_2 E_0^2 / 8\pi$: dividing by it removes one power. Counts (ii) and (iii) cancel, and what survives in (3.6) is exactly the ω^{-2} of the vertex. It is kinematics of the $\mathbf{A} \cdot \mathbf{p}$ form, not a density-of-states effect. Cross-check: in the length gauge the same factor hides inside the dipole matrix element, $|\mathbf{p}_{ul}|^2 = m^2 \omega_{ul}^2 |\mathbf{r}_{ul}|^2$, so ε_2 written with $|\mathbf{r}_{ul}|^2$ has no explicit $1/\omega^2$ — on shell ($\omega = \omega_{ul}$) the two forms agree identically.*

Near a critical point the interband matrix element varies slowly; following GRW we freeze it and average over polarization, in their gradient convention:

$$|P|^2 \equiv |\langle u | \nabla | l \rangle|^2 \quad (\text{length}^{-2}), \quad \overline{|M_{ul}|^2} = \overline{|\hat{\boldsymbol{\varepsilon}} \cdot \mathbf{p}_{ul}|^2} = \frac{\hbar^2 |P|^2}{3}. \quad (3.7)$$

Per critical-point family this turns (3.6) into the working form

$$\varepsilon_2(\hbar\omega, T) = \frac{4\pi^2 e^2 \hbar^2}{3m^2 \omega^2} \cdot \frac{2}{(2\pi)^3} N |P|^2 \int_{\text{half-nbhd}} d^3k \delta(\Omega_{lu}(\mathbf{k}) - \hbar\omega) [1 - f(E)] \Big|_{E=\hbar\omega_u(\mathbf{k})}, \quad (3.8)$$

with $\Omega_{lu} \equiv \hbar\omega_u - \hbar\omega_l$ and N the half-neighborhood count of Sec. 2.

3.5 Occupation factor: Rosei's $[1 - f]$

The exact net weight of (3.6) — absorption minus stimulated emission — is the difference $f(E - \hbar\omega) - f(E)$ read at the final-state energy $E = \hbar\omega_u(\mathbf{k})$. For the noble-metal d bands the initial state sits at $E - \hbar\omega \lesssim -1.5 \text{ eV} \ll -k_B T$, so $f(E - \hbar\omega) = 1$ to machine precision and the difference is Rosei's factor,

$$f(E - \hbar\omega) - f(E) \longrightarrow [1 - f(E, T)], \quad (3.9)$$

which is used throughout this document, exactly as GRW use it. (The non-equilibrium extension puts f^S or f^P into the same slot; the *emission* weight is a different object — a product, not a difference — and is derived in Sec. 7.)

4 Reduction of the k-integral: the common machinery

4.1 The Rosei two-band model

Both critical points are axially symmetric parabolic pairs. With the positive levels/masses of Sec. 2, generic levels $\hbar\omega_{u0} > 0$ (empty, above E_F) and $\hbar\omega_{l0} > 0$ (occupied d level below E_F), and the *topology switch* σ_u ,

$$\boxed{\begin{aligned} E &= \hbar\omega_u(\mathbf{k}) = +\hbar\omega_{u0} + A_u k_\perp^2 + \sigma_u B_u k_\parallel^2, & \sigma_u^X &= -1 \text{ (saddle),} \\ E - \hbar\omega &= \hbar\omega_l(\mathbf{k}) = -\hbar\omega_{l0} - A_l k_\perp^2 - B_l k_\parallel^2, & \sigma_u^L &= +1 \text{ (minimum),} \end{aligned}} \quad (4.1)$$

where $(\hbar\omega_{u0}, \hbar\omega_{l0}) = (\hbar\omega_{X_6^-}, \hbar\omega_{X_7^+})$ at X and $(\hbar\omega_{L_6^-}, \hbar\omega_{L^+})$ at L ; the lower band is a local maximum in both directions at both points. The vertical transition energy is

$$\Omega_{lu}(\mathbf{k}) = \mathcal{E}_g + \bar{A} k_\perp^2 + \bar{B} k_\parallel^2, \quad \mathcal{E}_g = \hbar\omega_{u0} + \hbar\omega_{l0}, \quad \bar{A} = A_u + A_l > 0, \quad \boxed{\bar{B} = B_l + \sigma_u B_u} \quad (4.2)$$

— the entire topological difference between X and L is carried by σ_u through $\bar{B}$ (and the determinant $\mathcal{D}$ below). The gold parameters (GRW’s levels; C&S masses) are collected in the tables of Sec. 6.

4.2 Linearization $(u, v) = (k_\perp^2, k_\parallel^2)$

This is the step where the present derivation simplifies Rosei’s geometric construction. On the quadrant $u, v \geq 0$ the two constraints — fixing the final energy E and the transition energy $\Omega_{lu} = \hbar\omega$ — are *linear*:

$$\begin{pmatrix} E - \hbar\omega_{u0} \\ \hbar\omega - \mathcal{E}_g \end{pmatrix} = \underbrace{\begin{pmatrix} A_u & \sigma_u B_u \\ \bar{A} & \bar{B} \end{pmatrix}}_M \begin{pmatrix} u \\ v \end{pmatrix}, \quad \mathcal{D} \equiv \det M = A_u B_l - \sigma_u A_l B_u. \quad (4.3)$$

Stacking the two constraints *is* the construction of the transition matrix M ; its determinant is the Jacobian that will collapse the two δ -functions below. Inverting,

$$\boxed{u = \frac{\bar{B}(E - \hbar\omega_{u0}) - \sigma_u B_u(\hbar\omega - \mathcal{E}_g)}{\mathcal{D}}, \quad v = k_\parallel^2 = \frac{A_u(\hbar\omega - \mathcal{E}_g) - \bar{A}(E - \hbar\omega_{u0})}{\mathcal{D}}} \quad (4.4)$$

and the physical window is exactly $\{u \geq 0\} \cap \{v \geq 0\}$. The two point-specific determinants:

$$\mathcal{D}_X = A_u B_l + A_l B_u > 0 \text{ always (either sign of } \bar{B}_X), \quad \mathcal{D}_L = A_u B_l - A_l B_u (< 0 \text{ for Au).} \quad (4.5)$$

4.3 The energy-distributed JDOS: GRW’s Eqs. (4) and (5)

The occupation in (3.8) depends on $\mathbf{k}$ only through the final energy $E = \hbar\omega_u(\mathbf{k})$ — that is where temperature enters — so before any geometry we *sort the transitions by E* , by slipping the identity $1 = \int dE \delta(\hbar\omega_u(\mathbf{k}) - E)$ into (3.8):

$$\int_{k_\parallel > 0} d^3k \delta(\Omega_{lu} - \hbar\omega) [1 - f] = \int dE [1 - f(E)] \underbrace{\int_{k_\parallel > 0} d^3k \delta(\hbar\omega_u - E) \delta(\Omega_{lu} - \hbar\omega)}_{(2\pi)^3 D_{l \rightarrow u}(E, \hbar\omega)}. \quad (4.6)$$

The inner integral is Rosei’s central object, the *energy-distributed joint density of states*: transitions per unit crystal volume, per unit final energy, per unit transition energy (one half-neighborhood, one spin). Both δ ’s are accounted for: energy conservation was already in (3.8); the second is the inserted identity, and it is what later lets $\mathcal{J}$ weight each E by its own occupation.

Reduce it all the way — this is where $\mathcal{D}$ and $\mathcal{F}$ come from. Axial symmetry gives $d^3k = 2\pi k_\perp dk_\perp dk_\parallel$; in the squared variables $k_\perp dk_\perp = \frac{1}{2}du$, $dk_\parallel = dv/(2\sqrt{v})$:

$$\frac{1}{(2\pi)^3} \int_{k_\parallel > 0} d^3k = \frac{1}{(2\pi)^3} \frac{\pi}{2} \int_0^\infty du \int_0^\infty \frac{dv}{\sqrt{v}} = \frac{1}{16\pi^2} \int_0^\infty du \int_0^\infty \frac{dv}{\sqrt{v}}. \quad (4.7)$$

The two δ 's are exactly the two rows of M in (4.3); linear δ 's collapse onto the unique solution (4.4) divided by $|\det M| = |\mathcal{D}|$, and $\sqrt{v_*} = k_\parallel(E, \hbar\omega)$:

$$D_{l \rightarrow u}(E, \hbar\omega) = \frac{1}{16\pi^2 |\mathcal{D}|} \frac{1}{k_\parallel(E, \hbar\omega)}. \quad (4.8)$$

This is the whole result of the reduction: it contains only $\mathcal{D}$ and $k_\parallel$ — no $\mathcal{F}$ yet. Rosei's $\mathcal{F}_{l \rightarrow u}$ is not an independent input but a repackaging of $\mathcal{D}$, manufactured here. Expanding $\mathcal{D}_X$ in the masses,

$$\mathcal{D}_X = A_u B_l + A_l B_u = \left(\frac{\hbar^2}{2}\right)^2 \frac{m_{l\perp} m_{u\parallel} + m_{l\parallel} m_{u\perp}}{m_{l\perp} m_{l\parallel} m_{u\perp} m_{u\parallel}}, \quad (4.9)$$

and defining $\mathcal{F}_{l \rightarrow u} \equiv \hbar^2/(2\sqrt{|\mathcal{D}|})$ as pure shorthand, the $\hbar^2/2$ cancels and GRW's Eq. (5) falls out — derived, not assumed — together with its L analogue:

$$\mathcal{F}_X = \left[\frac{m_{l\perp} m_{u\parallel} + m_{l\parallel} m_{u\perp}}{m_{l\perp} m_{l\parallel} m_{u\perp} m_{u\parallel}} \right]^{-1/2}, \quad \mathcal{F}_L = \left[\frac{|m_{l\perp} m_{u\parallel} - m_{l\parallel} m_{u\perp}|}{m_{l\perp} m_{l\parallel} m_{u\perp} m_{u\parallel}} \right]^{-1/2} \quad (5)$$

(the sum/difference structure is the $\sigma_u = \mp 1$ dichotomy of (4.5)). Reading the shorthand backwards, $|\mathcal{D}| = \hbar^4/4\mathcal{F}^2$, and (4.8) becomes the EDJDOS in Rosei's own symbols:

$$\boxed{D_{l \rightarrow u}(E, \hbar\omega) = \frac{1}{16\pi^2 |\mathcal{D}|} \frac{1}{k_\parallel(E, \hbar\omega)} = \frac{\mathcal{F}_{l \rightarrow u}^2}{4\pi^2 \hbar^4} \frac{1}{k_\parallel(E, \hbar\omega)}} \quad (4')$$

supported on $u, v \geq 0$, with $k_\parallel$ from (4.4).

Where the printed (4) falls. GRW print $D_{l \rightarrow u} = (8\pi^2 \hbar^2)^{-1} \mathcal{F}/k_\parallel$. With $|P|^2$ the squared gradient matrix element of (3.7), the pair {printed (4), printed (9)} does not balance dimensionally, whereas (4') — forced, with nothing left to choose, by the definition of $D_{l \rightarrow u}$ in (4.6) — closes the FGR chain exactly through to ε_2 (Sec. 6). The two forms differ only by the constant $2\mathcal{F}/\hbar^2$ per critical point; a constant cannot bend a line shape, and it is absorbed into the fitted strength $S = \mathcal{F}|P|^2$ of Eq. (10) — which is why the slip is invisible in the published numbers. Details in App. A.

4.4 The thermally weighted JDOS: GRW's Eq. (7)

Carrying the occupation of (4.6) against $D_{l \rightarrow u}$ gives, verbatim as GRW write it,

$$\boxed{\mathcal{J}_{l \rightarrow u}(\hbar\omega, T) = \int_{E_{\min}}^{E_{\max}} D_{l \rightarrow u}(E, \hbar\omega) [1 - f(E, T)] dE} \quad (7)$$

with the windows $[E_{\min}, E_{\max}]$ set by the geometry ($u, v \geq 0$) and, where the geometry leaves an edge open, by the occupation itself — next section.

5 The two critical points, side by side

Everything so far is common. The topology forks the derivation through two signs only: σ_u (shape of $\bar{B}$) and $\text{sign}(\mathcal{D})$ (which window edge is singular). The X column follows GRW's own branch, $\bar{B}_X < 0$ (see the mass fork, Sec. 5.1); their equations then pop out directly.

X point ($\sigma_u = -1$)

Bands — exactly GRW's Eqs. (1)–(2): the X_6^- (sp) band curves up in the face, down along Δ (a saddle); X_7^+ (d) is a maximum.

$$E = \hbar\omega_{X_6^-} + \frac{\hbar^2 k_\perp^2}{2m_{u\perp}} - \frac{\hbar^2 k_\parallel^2}{2m_{u\parallel}}, \quad (1)$$

$$E - \hbar\omega = -\hbar\omega_{X_7^+} - \frac{\hbar^2 k_\perp^2}{2m_{l\perp}} - \frac{\hbar^2 k_\parallel^2}{2m_{l\parallel}}. \quad (2)$$

Vertical transitions live on the CEDS

$$\Omega_{lu}(\mathbf{k}) - \hbar\omega = 0. \quad (3)$$

Transition surface.

$$\Omega_{lu} = \mathcal{E}_g^X + \bar{A}_X k_\perp^2 + \bar{B}_X k_\parallel^2, \quad \bar{B}_X = B_l - B_u < 0$$

on GRW's branch ($m_{u\parallel} < m_{l\parallel}$: the sp band falls along Δ faster than the d band). The CEDS is an *open* hyperboloid at every $\hbar\omega$: transitions exist on both sides of $\mathcal{E}_g^X$ — a real sub-gap tail.

Determinant.

$$\mathcal{D}_X = A_u B_l + A_l B_u > 0 \quad (\text{either branch}).$$

Slaved $k_\parallel$ — GRW's Eq. (6), corrected (App. A):

$$k_\parallel^2 = \frac{2\mathcal{F}_X^2}{\hbar^2} \left[\frac{\hbar\omega - \hbar\omega_{X_7^+} - E}{m_{u\perp}} + \frac{\hbar\omega_{X_6^-} - E}{m_{l\perp}} \right]. \quad (6')$$

Window. E decreases without bound along the CEDS ($dE \propto -\mathcal{D}_X dv < 0$): no geometric floor. The integral is cut by occupation alone — GRW's practical floor $E_{\min} = -20k_B T$. The ceiling is geometric and changes character at the gap:

$$E_{\max} = \hbar\omega_{X_6^-} + \begin{cases} \frac{A_u}{\bar{A}_X}(\hbar\omega - \mathcal{E}_g^X), & \hbar\omega \geq \mathcal{E}_g^X, \\ -\frac{B_u}{|\bar{B}_X|}(\mathcal{E}_g^X - \hbar\omega), & \hbar\omega < \mathcal{E}_g^X, \end{cases} \quad (8')$$

the lower line being GRW's Eq. (8), corrected (App. A).

Singular edge. Above the gap the ceiling is the $k_\parallel \rightarrow 0$ circle, where $D_{l \rightarrow u} \propto 1/k_\parallel$ diverges:

$$\frac{1}{k_\parallel(E)} = \sqrt{\frac{\mathcal{D}_X}{\bar{A}_X}} \frac{1}{\sqrt{E_{\max} - E}}.$$

Below the gap the ceiling is the finite $k_\perp = 0$ point: no divergence.

L point ($\sigma_u = +1$)

Bands — not derived in GRW 1975 (borrowed from Rosei 1974); rederived here with the same tools. L_6^- is a *minimum* in both directions; L^+ (d top) a maximum.

$$E = \hbar\omega_{L_6^-} + \frac{\hbar^2 k_\perp^2}{2m_{u\perp}} + \frac{\hbar^2 k_\parallel^2}{2m_{u\parallel}},$$

$$E - \hbar\omega = -\hbar\omega_{L^+} - \frac{\hbar^2 k_\perp^2}{2m_{l\perp}} - \frac{\hbar^2 k_\parallel^2}{2m_{l\parallel}}.$$

The only change from X is the sign of the upper $k_\parallel^2$ term; that one flip cascades through everything below.

Transition surface.

$$\Omega_{lu} = \mathcal{E}_g^L + \bar{A}_L k_\perp^2 + \bar{B}_L k_\parallel^2, \quad \bar{B}_L = B_l + B_u > 0$$

unconditionally: a closed ellipsoid, existing only for $\hbar\omega \geq \mathcal{E}_g^L$. *No sub-gap tail at L* — the sharp-vs-tailed contrast of the two edges is baked in right here.

Determinant.

$$\mathcal{D}_L = A_u B_l - A_l B_u < 0 \quad \text{for Au}$$

$$(m_{l\perp} m_{u\parallel} = 0.084 < m_{l\parallel} m_{u\perp} = 0.247).$$

Slaved $k_\parallel$ — same inversion, $\mathcal{D}_L < 0$ reversing both inequalities:

$$k_\parallel^2 = \frac{2\mathcal{F}_L^2}{\hbar^2} \left[\frac{E - (\hbar\omega - \hbar\omega_{L^+})}{m_{u\perp}} + \frac{E - \hbar\omega_{L_6^-}}{m_{l\perp}} \right].$$

Window. Both edges geometric ($v \geq 0$ now gives the *lower* edge, $u \geq 0$ the upper):

$$E_{\min} = \hbar\omega_{L_6^-} + \frac{A_u}{\bar{A}_L}(\hbar\omega - \mathcal{E}_g^L),$$

$$E_{\max} = \hbar\omega_{L_6^-} + \frac{B_u}{\bar{B}_L}(\hbar\omega - \mathcal{E}_g^L),$$

slopes 0.745 and 0.896 for Au. The whole window rides upward from $\hbar\omega_{L_6^-} = 0.89$ eV (Remark 5.1).

Singular edge. From (4.4) with $\mathcal{D}_L < 0$,

$$k_\parallel^2 = \frac{\bar{A}_L}{|\mathcal{D}_L|} (E - E_{\min}),$$

so $D_{l \rightarrow u} \propto 1/k_\parallel$ diverges at the *lower* limit:

$$\frac{1}{k_\parallel(E)} = \sqrt{\frac{|\mathcal{D}_L|}{\bar{A}_L}} \frac{1}{\sqrt{E - E_{\min}}}.$$

Onset. At $\hbar\omega = \mathcal{E}_g^L$ exactly, with $\mathcal{J}_L \propto \sqrt{\hbar\omega - \mathcal{E}_g^L}$; the inverse-square-root weight sits on the onset edge itself, which is why the L edge is sharp and strong.

Onset. At $T = 0$ absorption starts when $E_{\max}$ crosses E_F :

$$\hbar\omega_{\text{on}} = \mathcal{E}_g^X - \hbar\omega_{X_6^-} \frac{m_{l\parallel} - m_{u\parallel}}{m_{l\parallel}} \simeq 1.83 \text{ eV},$$

the piezomodulation threshold; the tail 1.83–1.94 eV is purely geometric, and the onset is *step-like* (GRW’s word), not divergent.

Punchline of the comparison: the $k_{\parallel} \rightarrow 0$ singularity of $D_{l \rightarrow u}$ sits at the *upper* window edge at X and the *lower* edge at L — purely because $\mathcal{D}_X > 0$ while $\mathcal{D}_L < 0$; nothing physical was added. L is sharp and strong, X soft and tailed: Rosei’s “splitting of the interband absorption edge,” both halves from one formula set.

Remark 5.1 (The L level and the Fermi-surface neck) *GRW’s qualitative discussion attributes the strength of the L singularity to the onset CEDS being tangent to the Fermi surface along the neck — which requires L_6^- below E_F . The fitted scheme ($\mathcal{E}_g^L = 2.45$, $\hbar\omega_{L^+} = 1.56 \Rightarrow \hbar\omega_{L_6^-} = +0.89 \text{ eV}$) puts the whole L window above E_F , making the L channel inert through the Fermi factor. Both statements cannot hold in one parabolic model; the fitted scheme is what the code implements, and the tension is recorded here rather than papered over.*

5.1 The X mass fork: sign of $\bar{B}_X$

The one assignment at X I am not sure of, flagged because it flips a sign. My own extraction of the C&S Fig. 5 curvatures gives $m_{u\parallel} = 0.40$, $m_{l\parallel} = 0.15$ — the *opposite* $\parallel$ ordering to GRW’s, hence $\bar{B}_X > 0$: a *closed* window $[\hbar\omega_{X_6^-} - \frac{B_u}{B_X}(\hbar\omega - \mathcal{E}_g^X), E_{\max}]$, no geometric sub-gap tail, and a $\sqrt{\hbar\omega - \mathcal{E}_g^X}$ onset sitting exactly at the gap (the observed tail below 1.94 eV would then be broadening only, and GRW’s $-20k_B T$ floor would overshoot the true geometric floor). Three facts keep this fork contained:

- Above the gap both branches give the *same* line shape and the same singular edge ($8'$, upper line): the $\parallel$ masses enter only through $\mathcal{F}_X$, which the fit absorbs into $S_X = \mathcal{F}_X |P_X|^2$. An optical fit alone cannot distinguish the branches; only the sub-gap physics and the extracted $|P_X|^2$ can. (The $\perp$ assignment does move the above-gap slope $A_u/\bar{A}_X$: 0.62 on the GRW assignment of Sec. 6, 0.38 on mine; $\mathcal{F}_X$ is invariant under the full $u \leftrightarrow l$ swap.)
- Both orderings make the d band the lighter one along Δ , against the flat- d prior — a transcription swap in my own mass table is the likeliest fix.
- Until C&S Fig. 5 is re-measured, this document follows GRW ($\bar{B}_X < 0$) and quotes the branch wherever it matters.

6 Assembled ε_2 : GRW’s Eqs. (9)–(10) and the scorecard

Insert the EDJDOS ($4'$) into the FGR form (3.8). The bridge is the definition of $D_{l \rightarrow u}$ itself, Eq. (4.6): the $(2\pi)^3$ cancels, and with spin (the explicit 2) and the half-neighborhood counts $N_X = 6$, $N_L = 8$,

$$\varepsilon_2(\hbar\omega, T) = \frac{4\pi^2 e^2 \hbar^2}{3m^2 \omega^2} \cdot 2 \cdot \sum_{i=X,L} N_i |P_i|^2 \mathcal{J}_i(\hbar\omega, T), \quad (6.1)$$

which tidies to Rosei’s second headline equation, with no leftover constant:

$$\boxed{\varepsilon_2(\hbar\omega, T) = \frac{8\pi^2 e^2 \hbar^4}{3m^2 (\hbar\omega)^2} \sum_{i=X,L} N_i |P_i|^2 \mathcal{J}_i(\hbar\omega, T)} \quad (9)$$

(GRW leave the N_i implicit, so their fitted $|P_i|^2$ really means $N_i|P_i|^2$ up to normalization — harmless for the quoted ratio). That clean landing needs three things at once: $|P|^2$ is the squared gradient element (3.7); $D_{l \rightarrow u}$ is (4'), not the printed (4); spin and N_i are kept where shown. The quantities the data actually fix are the strengths

$$S_X = \mathcal{F}_X |P_X|^2, \quad S_L = \mathcal{F}_L |P_L|^2, \quad (10)$$

with $|P_X/P_L|^2 = 0.370$ from the fit to Johnson–Christy; under the corrected (4') the fitted combination maps onto the printed S_i through exactly the constant $2\mathcal{F}_i/\hbar^2$ of Sec. 4.3 — every published number keeps its meaning.

Fully substituted — the explicit integral a computer evaluates — (4') into (9), the $\hbar^4$'s cancelling:

$$\varepsilon_2(\hbar\omega, T) = \frac{2e^2}{3m^2(\hbar\omega)^2} \sum_{i=X,L} N_i \mathcal{F}_i^2 |P_i|^2 \int_{E_{\min}^{(i)}}^{E_{\max}^{(i)}} \frac{1 - f(E, T)}{k_{\parallel}^{(i)}(E, \hbar\omega)} dE \quad (11)$$

with $k_{\parallel}^X$ from (6'), $k_{\parallel}^L$ from its L twin (same bracket, every slot sign-reversed), and windows:

	$E_{\min}$	$E_{\max}$	edge behaviour
$X, \hbar\omega \geq \mathcal{E}_g^X$	$-20k_B T$ (occupation)	$\hbar\omega_{X_6^-} + \frac{A_u}{A_X}(\hbar\omega - \mathcal{E}_g^X)$	$1/\sqrt{}$ at $E_{\max}$
$X, \hbar\omega < \mathcal{E}_g^X$	$-20k_B T$ (occupation)	sub-gap line of (8')	finite everywhere
$L, \hbar\omega \geq \mathcal{E}_g^L$	$\hbar\omega_{L_6^-} + \frac{A_u}{A_L}(\hbar\omega - \mathcal{E}_g^L)$	$\hbar\omega_{L_6^-} + \frac{B_u}{B_L}(\hbar\omega - \mathcal{E}_g^L)$	$1/\sqrt{}$ at $E_{\min}$
$L, \hbar\omega < \mathcal{E}_g^L$	—	—	no states

Parameters (levels: GRW's fit; masses: C&S, the X row on GRW's branch, subject to Sec. 5.1; $\mathcal{F}$ in units of the free-electron mass m ; $\mathcal{D}$ in units $(\hbar^2/2m)^2$):

	$\mathcal{E}_g$ (eV)	upper level (eV)	lower level (eV)	$m_{u\perp}$	$m_{u\parallel}$	$m_{l\perp}$	$m_{l\parallel}$
X	1.94	$\hbar\omega_{X_6^-} = 0.17$	$\hbar\omega_{X_7^+} = 1.77$	0.19	0.15	0.31	0.40
L	2.45	$\hbar\omega_{L_6^-} = 0.89$	$\hbar\omega_{L_7^+} = 1.56$	0.24	0.12	0.70	1.03

	$\mathcal{D}$	$\mathcal{F}/m$	window slopes	singular edge
X	+34.7	0.170	$E_{\max}$ (above gap): 0.62	upper ($k_{\parallel} \rightarrow 0$)
L	-7.86	0.357	$E_{\min}$: 0.745, $E_{\max}$: 0.896	lower ($k_{\parallel} \rightarrow 0$)

Scorecard — the ten GRW equations, one by one:

GRW	here	status	issue as printed
(1)	(1)	recreated	—
(2)	(2)	recreated	—
(3)	(3)	recreated	—
(4)	(4')	suggested fix	prefactor $(8\pi^2\hbar^2)^{-1}\mathcal{F} \rightarrow (4\pi^2\hbar^4)^{-1}\mathcal{F}^2$; off by const. $2\mathcal{F}/\hbar^2$
(5)	(5)	recreated	—
(6)	(6')	suggested fix	dimensionally impossible as printed
(7)	(7)	recreated	—
(8)	(8')	suggested fix	$m_{u\parallel} \leftrightarrow m_{l\parallel}$ subscripts swapped
(9)	(9)	recreated	N_i made explicit
(10)	(10)	recreated	absorbs the (4) constant; see App. B

7 Photonic density of states and the emission counterpart

7.1 Mode counting

Periodic quantization in volume V : two transverse polarizations, $\omega = cq$:

$$\rho(\omega) d\omega = \frac{2}{V} \frac{V}{(2\pi)^3} 4\pi q^2 dq \implies \boxed{\rho(\omega) = \frac{\omega^2}{\pi^2 c^3}} \quad (7.1)$$

modes per unit volume per unit angular frequency. In a transparent host of index n , $c \rightarrow c/n$; in a structured environment $\rho \rightarrow \rho(\mathbf{r}, \omega)$, the *local* density of states (Novotny & Hecht) — precisely the photonic factor of the PL factorization used in this thesis.

7.2 Spontaneous emission rate of one pair

FGR with the quantized field (3.3) at $n_q = 0$, summed over modes, using $\sum_\lambda \int d\Omega_q |\hat{\epsilon}_\lambda \cdot \mathbf{p}_{ul}|^2 = \frac{8\pi}{3} |\mathbf{p}_{ul}|^2$ and $|\mathbf{p}_{ul}|^2 = \hbar^2 |P|^2$:

$$\boxed{W_{\text{sp}}(\omega) = \frac{4e^2 \omega \hbar |P|^2}{3m^2 c^3} = \underbrace{\frac{4\pi^2 e^2 \hbar |P|^2}{3m^2 \omega}}_{\text{matter}} \times \underbrace{\frac{\omega^2}{\pi^2 c^3}}_{\rho(\omega)}} \quad (7.2)$$

— the vacuum Einstein- A rate in the $\mathbf{A} \cdot \mathbf{p}$ form. The factorization into a matter part times $\rho(\omega)$ is exact and is the microscopic origin of the LDOS $\times$ electronic split assumed in Chapter 2.

7.3 The interband PL spectrum and the Kirchhoff check

The emission weight is not a difference but a *product*: the upper level must be filled and the lower empty, $f(E)[1 - f(E - \hbar\omega)]$. Summing (7.2) over pairs (spin 2, N half-neighborhoods, same $D_{l \rightarrow u}$ and windows as in absorption):

$$\boxed{R(\hbar\omega) = \rho(\omega) \frac{4\pi^2 e^2 \hbar |P|^2}{3m^2 \omega} 2N \int dE f(E)[1 - f(E - \hbar\omega)] D_{l \rightarrow u}(E, \hbar\omega)} \quad (7.3)$$

photons per unit time, volume and photon energy. In equilibrium the Fermi identity $[1 - f]/f = e^{(E - \mu)/k_B T}$, applied at both levels, collapses the product (detailed balance; only the *transition* energy survives):

$$f(E)[1 - f(E - \hbar\omega)] = n_B(\hbar\omega) [f(E - \hbar\omega) - f(E)], \quad n_B(x) = \frac{1}{e^{x/k_B T} - 1}, \quad (7.4)$$

and since $[f(E - \hbar\omega) - f(E)]$ is Rosei's $[1 - f(E)]$ in the same deep- d -band limit used throughout (Sec. 3.5), every electronic factor of (7.3) maps onto (11) and the constants collapse *exactly*:

$$\boxed{R(\hbar\omega) = \frac{\omega^3}{\pi^2 \hbar c^3} \varepsilon_2(\omega) n_B(\hbar\omega) = \frac{\omega \rho(\omega)}{\hbar} \varepsilon_2(\omega) n_B(\hbar\omega)} \quad (7.5)$$

— the van Roosbroeck–Shockley relation; per thermal photon, emission over absorption is $e^{-\hbar\omega/k_B T}$: Kirchhoff. That (9) and (7.3), derived independently, satisfy (7.5) with no leftover constant is the strongest global check that every prefactor in this document is right.

Out of equilibrium (f^S, f^P) the collapse (7.4) fails — it used one shared (μ, T) at both levels — and the ratio $n_{\text{eff}}(E, \hbar\omega) = f(E)[1 - f(E - \hbar\omega)]/[f(E - \hbar\omega) - f(E)]$ retains genuine E -dependence: the emission line shape parts company with ε_2 . That deviation *is* the non-equilibrium PL signal this thesis is after; reproducing $n_{\text{eff}} \rightarrow n_B$ under f_{FD} is a unit test, *taking* it is throwing the signal away. In a structured environment, $\rho(\omega) \rightarrow \rho(\mathbf{r}, \omega)$ in (7.3).

8 Numerics: evaluate in $k_{\parallel}$, not in E

Groundwork for the implementation. The $1/\sqrt{}$ edges of (11) would force adaptive quadrature or endpoint-weighted rules — but the divergence is a pure coordinate artifact, manufactured in Sec. 4.3 when the smooth $\mathbf{k}$ -space geometry was re-parametrized by final energy. One exact change of variable inside the same 1D integral undoes it. Take $k \equiv k_{\parallel}$ itself: E is affine in k^2 , so $dE = \mp(\mu_{i\perp} \hbar^2 / \mathcal{F}_i^2) k dk$ while $D_{l \rightarrow u} \propto 1/k$ — the factors of k cancel identically,

$$D_{l \rightarrow u} dE = \mp \frac{\mu_{i\perp}}{4\pi^2 \hbar^2} dk_{\parallel}, \quad (8.1)$$

i.e. $k_{\parallel}$ is (up to constants) the cumulative count of the EDJDOS: integrating in $k_{\parallel}$ is integrating in E with the known weight $D_{l \rightarrow u}$ absorbed exactly into the node placement. Eq. (7) flattens to

$$\boxed{\mathcal{J}_i(\hbar\omega, T) = \frac{\mu_{i\perp}}{4\pi^2 \hbar^2} \int_{k_1^{(i)}}^{k_2^{(i)}} [1 - f(E_i(k), T)] dk} \quad \frac{1}{\mu_{i\perp}} \equiv \frac{1}{m_{u\perp}} + \frac{1}{m_{l\perp}}, \quad (12)$$

$$E_i(k) = \underbrace{\hbar\omega_{i_6^-} + \frac{\mu_{i\perp}}{m_{u\perp}} (\hbar\omega - \mathcal{E}_g^i)}_{k_{\parallel}=0 \text{ intercept}} \mp \frac{\mu_{i\perp} \hbar^2}{2\mathcal{F}_i^2} k^2 \quad (- \text{ at } X, + \text{ at } L), \quad (8.2)$$

valid on both sides of the X gap — all the branch logic lives in the limits. These are the $u = 0/v = 0$ endpoints of Sec. 5, written out since an implementation touches them first. At X (GRW branch):

$$k_1^X = \begin{cases} 0, & \hbar\omega \geq \mathcal{E}_g^X, \\ \frac{1}{\hbar} \left[\frac{2(\mathcal{E}_g^X - \hbar\omega) m_{u\parallel} m_{l\parallel}}{m_{l\parallel} - m_{u\parallel}} \right]^{1/2}, & \hbar\omega < \mathcal{E}_g^X, \end{cases} \quad k_2^X = k_{\parallel}^X(E_{\text{floor}}, \hbar\omega) \text{ from (6')}, \quad (8.3)$$

with $E_{\text{floor}} = -20k_B T$ in equilibrium; for a pumped distribution take $E_{\text{floor}} = -(\hbar\omega_{\text{pump}} + 20k_B T)$ — generosity is cheap, since below the true floor the integrand is ≈ 0 . At L both limits are geometric:

$$k_1^L = 0, \quad k_2^L = \frac{1}{\hbar} \sqrt{2\mu_{L\parallel} (\hbar\omega - \mathcal{E}_g^L)}, \quad \frac{1}{\mu_{L\parallel}} \equiv \frac{1}{m_{u\parallel}} + \frac{1}{m_{l\parallel}}. \quad (8.4)$$

No singularity survives: the integrand of (12) is a bounded, smooth occupation profile whose only feature is the Fermi step of width $\sim k_B T$. **Recipe** (composite Simpson on a uniform grid is entirely adequate):

1. $k = \text{linspace}(k_1(\hbar\omega), k_2(\hbar\omega), N)$, N odd;
2. integrand $1 - f(E_i(k))$ — Fermi–Dirac in equilibrium, interpolation of the tabulated f^S/f^P otherwise (use a *monotone* interpolant, e.g. PCHIP, to avoid overshoot at the distribution’s kinks);
3. `simpson(y, x=k)`, times $\mu_{i\perp}/4\pi^2 \hbar^2$; then into (9).

Measured accuracy (equilibrium, $\hbar\omega = 2.4$ eV, against a converged reference): X reaches 5×10^{-10} relative error at $N = 101$ and machine precision by $N = 201$; the L integrand in equilibrium is constant across the window (Remark 5.1), and even a hot 2000 K distribution gives 3×10^{-14} at $N = 101$. By contrast, uniform-grid Simpson in E on a singular edge stalls near 10^{-2} even at $N = 6401$ ($O(\sqrt{h})$ decay). Energy space *done properly* — the substitution $t = \sqrt{E_{\text{edge}} - E}$, or Gauss–Jacobi with the $1/\sqrt{}$ weight — is bit-identical to (12), because $t \propto k_{\parallel}$ exactly: the geometry had already named the correct quadrature variable. Hence the division of labor: *derive and reason in E ; evaluate in $k_{\parallel}$* .

Practical points.

- *Resolve the step*: the only feature sits at $k_{\text{step}} = k(E=0)$, trivially located since $E_i(k)$ is affine in k^2 ; at low T either raise N or split the interval there and Simpson each panel.
- *Respect the kinks* of a pumped f^S : slope discontinuities at $E_F \pm \hbar\omega_{\text{pump}}$ are *distribution* features no variable change removes — put panel boundaries at their images $k(E_F \pm \hbar\omega_{\text{pump}})$.
- *Vectorize over $\hbar\omega$* : build $k = k_1[:, \text{None}] + (k_2 - k_1)[:, \text{None}] s[\text{None}, :]$ with s uniform on $[0, 1]$ and Simpson along the last axis — the whole spectrum in one call. Gauss–Legendre with 48–64 nodes does the same job with fewer points if node count matters.
- *Cache the geometry*: $k_{1,2}$, $E_i(k)$ and the grid depend on $(\hbar\omega)$ only — not on T or on the distribution — so precompute them once and sweep temperatures/pump powers cheaply (this is what makes fitting fast).
- *Error control*: recompute at $2N$ and compare (one doubling).
- *Emission*: the PL integrand (7.3) swaps $[1 - f] \rightarrow f(E)[1 - f(E - \hbar\omega)]$ in the same quadrature — geometry, windows and nodes unchanged.

Built-in checks. (i) *Empty-window arithmetic*: wherever the occupation factor is $\equiv 1$ across the window, $\mathcal{J}_i = \frac{\mu_{i\perp}}{4\pi^2\hbar^2}(k_2 - k_1)$ exactly; at L with $T \rightarrow 0$ this is the textbook M_0 onset $\mathcal{J}_L = \frac{\mu_{L\perp}}{4\pi^2\hbar^2} \sqrt{(\hbar\omega - \mathcal{E}_g^L)/\bar{B}_L}$ in closed form. (ii) *Two-forms agreement*: (11) by adaptive quadrature (endpoint declared) and (12) by Simpson must agree to quadrature accuracy at every $(\hbar\omega, T)$ — a one-line unit test. (iii) *Kirchhoff*: with $f = f_{\text{FD}}$, confirm $n_{\text{eff}}(E, \hbar\omega) \rightarrow n_B(\hbar\omega)$ independent of E (Sec. 7).

A Corrections to GRW’s printed Eqs. (4), (6) and (8)

Each failure stated simply, then fixed.

Eq. (4). Printed: $D_{l \rightarrow u} = (8\pi^2\hbar^2)^{-1} \mathcal{F}_{l \rightarrow u} k_{\parallel}^{-1}$. *Why it fails*: the definition of $D_{l \rightarrow u}$ (Eq. (4.6)) leaves nothing to choose — the reduction of Sec. 4.3 gives $(16\pi^2|\mathcal{D}|)^{-1} k_{\parallel}^{-1} = \mathcal{F}^2/(4\pi^2\hbar^4 k_{\parallel})$, Eq. (4’), and with the gradient convention for $|P|^2$ only (4’) makes $\{(4), (9)\}$ dimensionally consistent and lands (9) with no leftover constant. The printed form is off by

$$\frac{\text{printed (4)}}{\text{correct (4')}} = \frac{2\mathcal{F}_{l \rightarrow u}}{\hbar^2}, \quad (\text{A.1})$$

a pure constant per critical point — it cannot bend a line shape, and it is absorbed once and for all into the fitted $S = \mathcal{F}|P|^2$ of Eq. (10). That is why the slip is easy to miss and costs nothing physically; but the assembly only closes into (9) if (4’) is used.

Eq. (6). Printed:

$$k_{\parallel} = \left(\hbar\omega - \hbar\omega_{X_7^+} + \frac{\hbar^2}{2m_{l\perp}}(\hbar\omega_{X_6^-} - E) - \frac{\hbar^2}{2m_{u\perp}}E \right)^{1/2}. \quad (6, \text{ as printed})$$

Why it fails: a wavevector is set equal to the square root of a sum of *energies*, and inside the bracket a bare energy is added to $(\hbar^2/2m) \times \text{energy}$ terms — mismatched units throughout; the coefficients read $\hbar^2/2m$ where $2m/\hbar^2$ is needed for the bracket to come out as length^{-2} . Three small fixes turn it into (6’): square the left-hand side; invert the coefficients and pull out the overall $2\mathcal{F}^2/\hbar^2$; restore the missing $1/m_{u\perp}$ on the first term. The three numerator slots of

$$k_{\parallel}^2 = \frac{2\mathcal{F}_X^2}{\hbar^2} \left[\frac{\hbar\omega - \hbar\omega_{X_7^+} - E}{m_{u\perp}} + \frac{\hbar\omega_{X_6^-} - E}{m_{l\perp}} \right] = \frac{A_u(\hbar\omega - \mathcal{E}_g^X) - \bar{A}_X(E - \hbar\omega_{X_6^-})}{\mathcal{D}_X} \quad (6')$$

then line up with the print term by term — confirmation that (6') is what was meant: a typo, but a disabling one, since the printed formula is unusable as written.

Eq. (8). Printed: $E_{\max} = \hbar\omega_{X_6^-} + (\mathcal{E}_g^X - \hbar\omega) m_{u\parallel} / (m_{l\parallel} - m_{u\parallel})$. Solving the model for the $k_{\perp} = 0$ edge gives

$$E_{\max} = \hbar\omega_{X_6^-} + (\mathcal{E}_g^X - \hbar\omega) \frac{m_{l\parallel}}{m_{u\parallel} - m_{l\parallel}}, \quad (8')$$

i.e. the printed equation has $m_{u\parallel}$ and $m_{l\parallel}$ interchanged — one consistent swap, specific to (8): the subscripts of (6) are *not* swapped, so it is not the same slip repeated. *The tell:* on GRW's own branch $m_{u\parallel} < m_{l\parallel}$, the printed denominator is positive and sends $E_{\max}$ *above* $\hbar\omega_{X_6^-}$ as $\hbar\omega$ drops below the gap — but the sub-gap CEDS has no states up there. (8') puts the sub-gap edge sensibly *below* $\hbar\omega_{X_6^-}$, which is what makes the tail (and the 1.83 eV step onset) work. On the alternative branch of Sec. 5.1 ($\bar{B}_X > 0$), the same algebraic edge is instead the *lower* window limit.

B Dictionary of conventions across the repository

Document	upper band	lower band	$\bar{B}$	status
GRW 1975; this doc (X)	$+A_u k_{\perp}^2 - B_u k_{\parallel}^2$	$-A_l k_{\perp}^2 - B_l k_{\parallel}^2$	$B_l - B_u$	canonical
<i>Equivalence</i> note (X)	$+A_u k_{\perp}^2 + B_u k_{\parallel}^2$	$-A_l k_{\perp}^2 + B_l k_{\parallel}^2$	$B_u - B_l$	same $\mathcal{D}$ (sign-robust)
A8 / A8.L	$+A_c u - B_c v$	$-A_v u - B_v v$	$B_v - B_c$	A8 ok ^(a) ; A8.L wrong at $L^{(b)}$
<code>interband_derivation_v2</code>	mixed	mixed	—	internally inconsistent ^(c)

^(a) A8's $\mathcal{E}_{\min}$ denominator ($B_v + B_c$) is the $\sigma_u = +1$ formula; for its own X conventions it should be $\bar{B} = B_v - B_c$.

^(b) A8.L models the L conduction band as a saddle ($-B_c$); C&S give $m_{u\parallel}^L = +0.12$: it is a minimum, $\sigma_u = +1$. ^(c) v2 asserts $\bar{B} > 0$ *and* a geometric sub-gap tail (requires $\bar{B} < 0$), and labels L as M_1 .

Normalizations. Earlier repo documents use the full-neighborhood, no- $(2\pi)^3$ kernel $K(E, \hbar\omega) = \int d^3k \delta\delta$ and the half-line measure $\mathcal{J}_{\text{eq}}$ of the *Equivalence* note; with $D_{l \rightarrow u}$ from (4'),

$$K = 2(2\pi)^3 D_{l \rightarrow u}, \quad \mathcal{J}_{\text{eq}} = \frac{1}{2}K, \quad \text{printed (4)} = \frac{2\mathcal{F}}{\hbar^2} \times (4'). \quad (\text{B.1})$$

Matrix elements: Rosei's $|P|^2$ is the squared *gradient* element, related to the SI momentum element by $|\mathbf{p}_{ul}|^2 = \hbar^2 |P|^2$. With these two conventions — (4') and gradient $|P|^2$ — GRW's Eq. (9) is exact as printed; no residual constant needs to be hidden in $|P|^2$, and only the strengths $S = \mathcal{F}|P|^2$ of Eq. (10) are fixed by the data (the printed-(4) constant reshuffles $D_{l \rightarrow u}$, $\mathcal{F}$, $|P|^2$ among themselves without touching S).

C Dimensional audit (Gaussian)

$[e^2] = \text{erg cm}$, $[|P|^2] = \text{cm}^{-2}$, $[A_i] = [B_i] = \text{erg cm}^2$, $[\mathcal{D}] = \text{erg}^2 \text{cm}^4$, $[\mathcal{F}] = \text{g}$, $[D_{l \rightarrow u}] = \text{erg}^{-2} \text{cm}^{-3}$, $[\mathcal{J}] = \text{erg}^{-1} \text{cm}^{-3}$.

- (4'): $\mathcal{F}^2 / \hbar^4 k_{\parallel} \sim \text{g}^2 / (\text{erg}^4 \text{s}^4 \text{cm}^{-1}) = \text{erg}^{-2} \text{cm}^{-3}$ — states per volume per (energy)², as a double- δ measure must be. ✓
- (9): $\frac{\text{erg cm} \cdot \text{erg}^4 \text{s}^4}{\text{g}^2 \text{erg}^2} \cdot \text{cm}^{-2} \cdot \text{erg}^{-1} \text{cm}^{-3} = 1$ (using $\text{erg s}^2 = \text{g cm}^2$). ✓
- (12): $\mu_{\perp} / \hbar^2 \times [k] \sim \text{g cm}^{-1} / \text{erg}^2 \text{s}^2 = \text{erg}^{-1} \text{cm}^{-3} = [\mathcal{J}]$. ✓
- (7.1): $[\rho] = \text{s cm}^{-3}$. ✓
- (7.2): $\frac{\text{erg cm} \cdot \text{s}^{-1} \cdot \text{erg s} \cdot \text{cm}^{-2}}{\text{g}^2 \text{cm}^3 \text{s}^{-3}} = \text{erg}^2 \text{cm}^{-1} / (\text{g}^2 \text{cm}^3 \text{s}^{-3} \text{s}) = \text{s}^{-1}$. ✓

- (7.3): $[\rho] \cdot [\text{matter}] \cdot [\mathcal{J}] = \text{s cm}^{-3} \cdot \text{cm}^3 \text{s}^{-2} \cdot \text{erg}^{-1} \text{cm}^{-3} = \text{erg}^{-1} \text{cm}^{-3} \text{s}^{-1}$ — photons per time, volume and photon energy. ✓
- (7.5): $[\omega\rho/\hbar] = \text{erg}^{-1} \text{cm}^{-3} \text{s}^{-1}$, times dimensionless $\varepsilon_2 n_B$. ✓

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